

AL AIN INTL, United Arab Emirates

WMO: 412180

Lat: 24.262N

Lon: 55.609E

Elev: 869

StdP: 14.24

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	51.9	53.7	24.5	19.2	85.8	28.3	23.0	86.9	24.5	73.0	21.2	70.3	6.0	40	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.8	114.8	72.6	113.1	72.6	111.5	72.6	83.8	96.5	82.3	96.0	80.8	96.0	9.9	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.0	166.5	90.1	79.0	155.5	89.8	77.0	145.4	90.0	49.1	96.1	47.1	95.8	45.4	96.1	90.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.0	20.2	18.1	47.7	117.6	2.1	1.1	46.2	118.3	45.0	119.0	43.9	119.6	42.4	120.4		
(5)			42.1	86.0	2.8	2.0	40.1	87.4	38.4	88.6	36.9	89.7	34.8	91.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	84.6	65.9	69.9	75.9	84.7	93.0	97.3	99.5	99.4	94.4	87.2	77.2	69.4
	DBStd	12.72	4.26	5.85	6.42	6.13	4.54	3.67	3.41	3.05	3.24	3.28	4.04	4.17
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	68	40	16	2	0	0	0	0	0	0	0	0	10
	CDD50	12615	494	557	802	1041	1332	1418	1536	1530	1333	1154	816	602
	CDD65	7208	69	153	339	591	867	968	1071	1065	883	689	366	147
	CDH74	109210	446	1324	3698	8079	13913	16403	18620	18512	14300	9506	3421	988
CDH80	74535	40	423	1778	4901	9750	12138	14160	14049	10022	5648	1421	205	
Wind	WSAvg	8.8	7.8	9.0	9.7	9.5	9.3	9.1	9.3	9.5	9.0	8.3	7.6	7.5
Precipitation	PrecAvg	2.9	0.4	0.4	0.7	0.3	0.1	0.0	0.2	0.2	0.1	0.0	0.1	0.3
	PrecMax	6.7	2.8	1.5	2.9	2.4	0.1	0.4	2.4	0.5	0.9	0.3	1.1	2.6
	PrecMin	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0
	PrecStd	1.8	0.7	0.4	0.9	0.5	0.0	0.1	0.5	0.1	0.2	0.1	0.2	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	83.7	91.8	100.1	106.8	114.3	116.7	116.6	115.2	112.2	105.4	95.2	87.8
		MCWB	62.3	64.2	65.6	68.4	70.2	71.6	72.7	72.8	71.5	68.8	67.3	61.9
	2%	DB	80.8	88.2	96.4	104.2	111.0	114.5	114.7	113.4	109.6	102.5	91.9	84.5
		MCWB	61.9	63.4	64.7	67.3	70.2	71.5	73.5	73.1	71.4	68.2	66.1	62.9
	5%	DB	78.8	85.4	93.1	101.9	109.0	112.7	113.0	112.4	107.8	100.6	90.1	82.3
		MCWB	60.8	62.6	64.6	67.1	69.9	71.3	73.7	73.3	71.3	67.6	66.1	63.2
	10%	DB	76.7	82.1	89.7	98.7	106.2	110.8	111.2	110.5	105.8	98.5	87.9	79.9
		MCWB	60.0	61.9	63.0	66.1	69.2	71.3	73.8	73.8	71.2	67.8	65.9	62.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.5	70.1	72.6	75.3	78.7	83.7	86.2	85.7	83.7	79.4	74.6	70.0
		MCDB	76.0	81.0	86.6	94.0	96.6	96.8	98.6	96.6	92.9	86.1	83.1	78.4
	2%	WB	65.6	67.4	70.2	72.7	76.3	81.0	84.2	83.6	81.2	77.3	72.4	67.8
		MCDB	74.8	79.6	83.4	92.3	95.9	95.6	97.5	96.1	91.4	85.7	81.5	76.2
	5%	WB	64.0	65.6	68.3	70.8	74.7	79.4	82.6	81.8	79.7	75.7	70.9	66.4
		MCDB	73.7	78.9	81.8	90.8	95.9	95.2	97.1	96.8	91.0	85.7	80.6	76.1
	10%	WB	62.4	64.1	66.7	69.1	73.1	77.8	81.0	80.0	78.1	74.1	69.5	65.1
		MCDB	72.0	76.8	81.7	89.9	96.3	95.3	97.4	97.6	91.4	85.9	80.3	74.8
Mean Daily Temperature Range	5% DB	MDBR	19.3	21.1	22.8	24.3	26.4	26.7	23.8	22.8	23.5	22.7	20.3	19.7
		MCDBR	21.0	23.0	25.3	25.8	28.2	27.9	24.6	24.1	24.8	24.3	21.9	21.8
		MCWBR	8.8	9.4	9.5	10.0	9.8	10.0	9.6	9.6	10.5	11.4	8.8	8.6
	5% WB	MCDBR	17.9	19.5	20.8	23.2	27.1	27.0	24.7	23.9	24.8	23.7	21.2	18.7
		MCWBR	9.1	9.4	9.3	10.3	10.6	10.2	9.5	10.1	11.1	10.9	8.8	8.4
Clear-Sky Solar Irradiance	taub		0.417	0.470	0.534	0.571	0.569	0.630	0.719	0.688	0.585	0.481	0.428	0.410
	taud		2.058	1.900	1.743	1.694	1.694	1.602	1.495	1.542	1.697	1.903	2.052	2.080
	Ebn at Noon		262	256	246	240	239	223	204	210	229	249	256	258
	Edh at Noon		47	59	73	78	78	85	94	89	75	58	47	44
All-Sky Solar Radiation	RadAvg		1384	1630	1911	2149	2382	2408	2226	2181	2048	1808	1480	1313
	RadStd		74	103	115	123	54	60	80	51	37	33	32	66

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	+0.85	+1.26	N/A	N/A	N/A	N/A	N/A	N/A	+270	+268

Nomenclature: See separate page